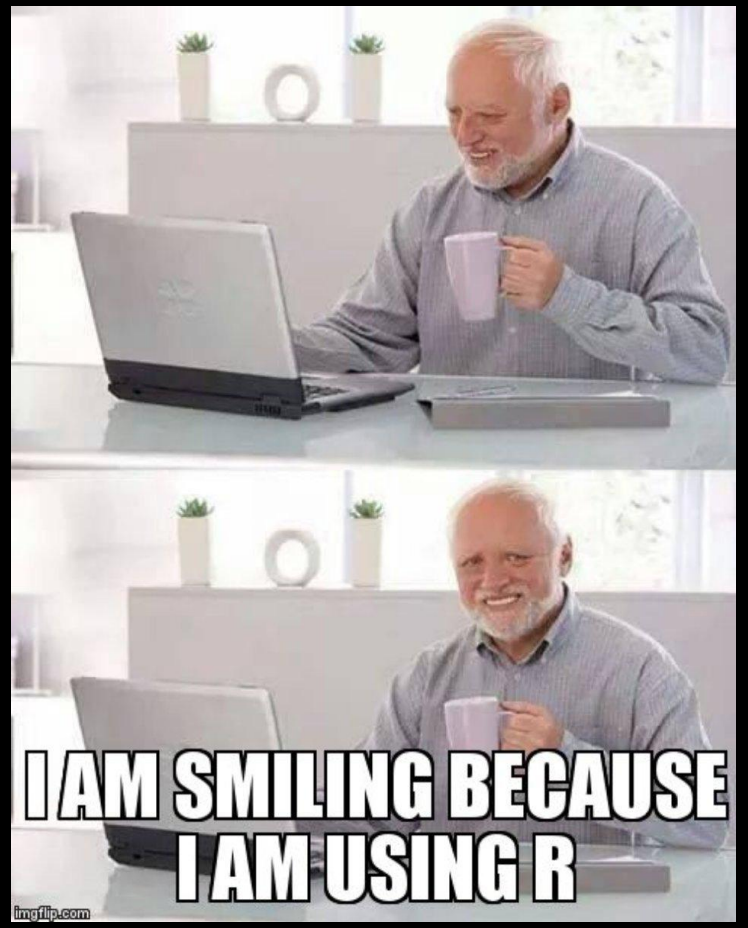


MES 503 Lab 5

Kayla Blincow, PhD

Announcements

- Lab assignment 2 posted on Brightspace
  - Due next Wednesday
- Quiz 2 next Monday
- Share papers with me please ☺

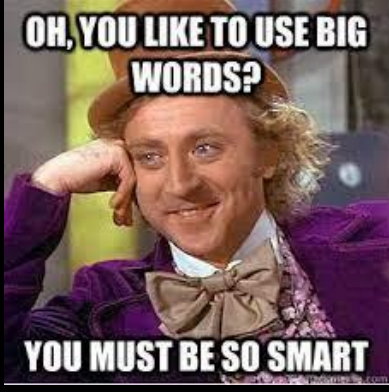
TENTATIVE SCHEDULE OF TOPICS – MES 503 Fall 2026

		Mondays (Lectures)		Wednesdays (Labs)	
Week	**Date**	**Lecture Topic**	**Date**	**Lab topic**	**Assessments**
1	8/17	Introduction to Statistics & Experimental Design I	8/19	Introduction to R and RStudio	
2	8/24	Introduction to Statistics & Experimental Design II	8/26	Intro to ggplot2	
3	8/31	Correlation	9/2	Correlation in R	Quiz 1
4	9/7	LABOR DAY (No Class)	9/9	Intro to Data wrangling	Lab 1 Due (9/10)
5	9/14	Statistical Hypothesis Testing & t tests	9/16	t Tests in R	
6	9/21	Linear Regression	9/23	Linear Regression in R	Quiz 2 (9/21) Lab 2 Due (9/24)
7	9/28	ANOVA	9/30	ANOVA in R	
8	10/5	ANOVA II (Two-Way & Nested) (virtual)	10/7	Two-Way & Nested ANOVA in R	Quiz 3
9	10/12	Multiple Linear Regression	10/14	Multiple Linear Regression in R	Lab 3 Due (10/15)
10	10/19	ANCOVA	10/21	ANCOVA in R	Quiz 4
11	10/26	Analysis of Frequencies	10/28	Analysis of Frequencies in R	Lab 4 Due (10/29)
12	11/2	Liberty Day – UVI Closed	11/4	Miscellaneous Fun R Topic	Quiz 5
13	11/9	Quantifying Variability/Figure and Data Sheet Design	11/11	Veterans Day – UVI Closed	
14	11/16	Reading for Stats Presentations	11/18	Reading for Stats Presentations	Lab 5 Due (11/17)
15	11/23	Applying this course to your thesis	11/25	Study Day	
16			12/3	**Final Exam**	

t-tests!

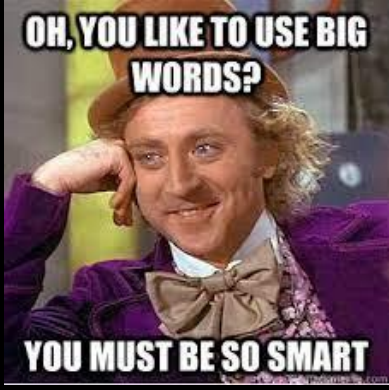
- When do we use them?
- What type of data is our explanatory variable?
- What type of data is our response variable?

t-Test



- Used to test for **differences between two means**
- Different types:
  - One sample
  - Two sample
  - One-tailed
  - Two-tailed
- Type depends on your question and hypothesis
- Null distribution – Student's t distribution
- Test statistic – t test statistic (what you're calculating to compare against the distribution)

t-Test Assumptions

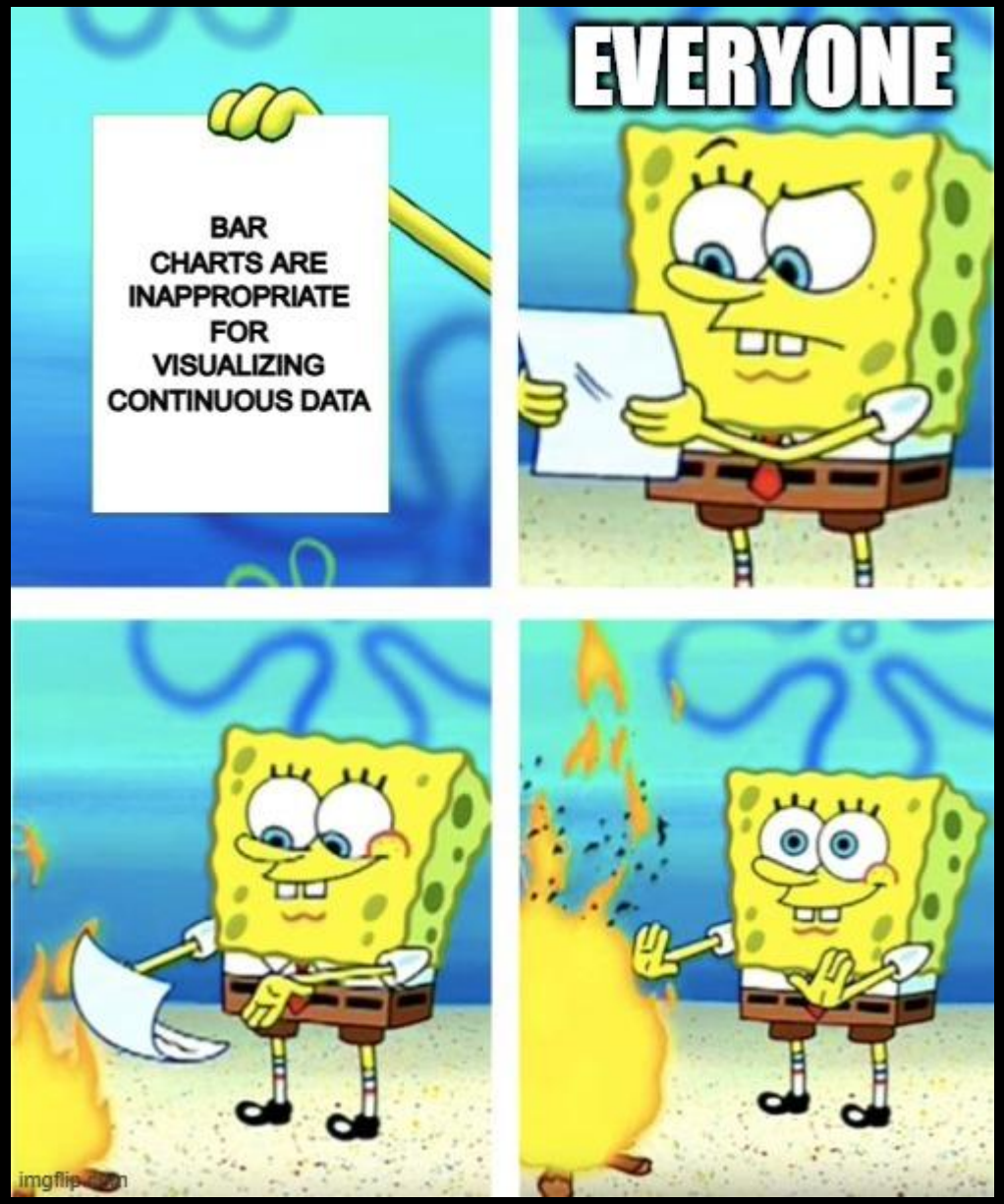


- Your sample means are normally distributed
- Depending on type:
  - Assumption of equal variance between sampled populations (exception two sample t-test for unequal variance)
  - Assumption of independence between sampled groups (exception paired t-tests)

- t tests are generally robust to violations of assumptions

Side quest...

BAR PLOTS ARE NOT MEANT FOR DISPLAYING CONTINUOUS DATA OR MEANS.



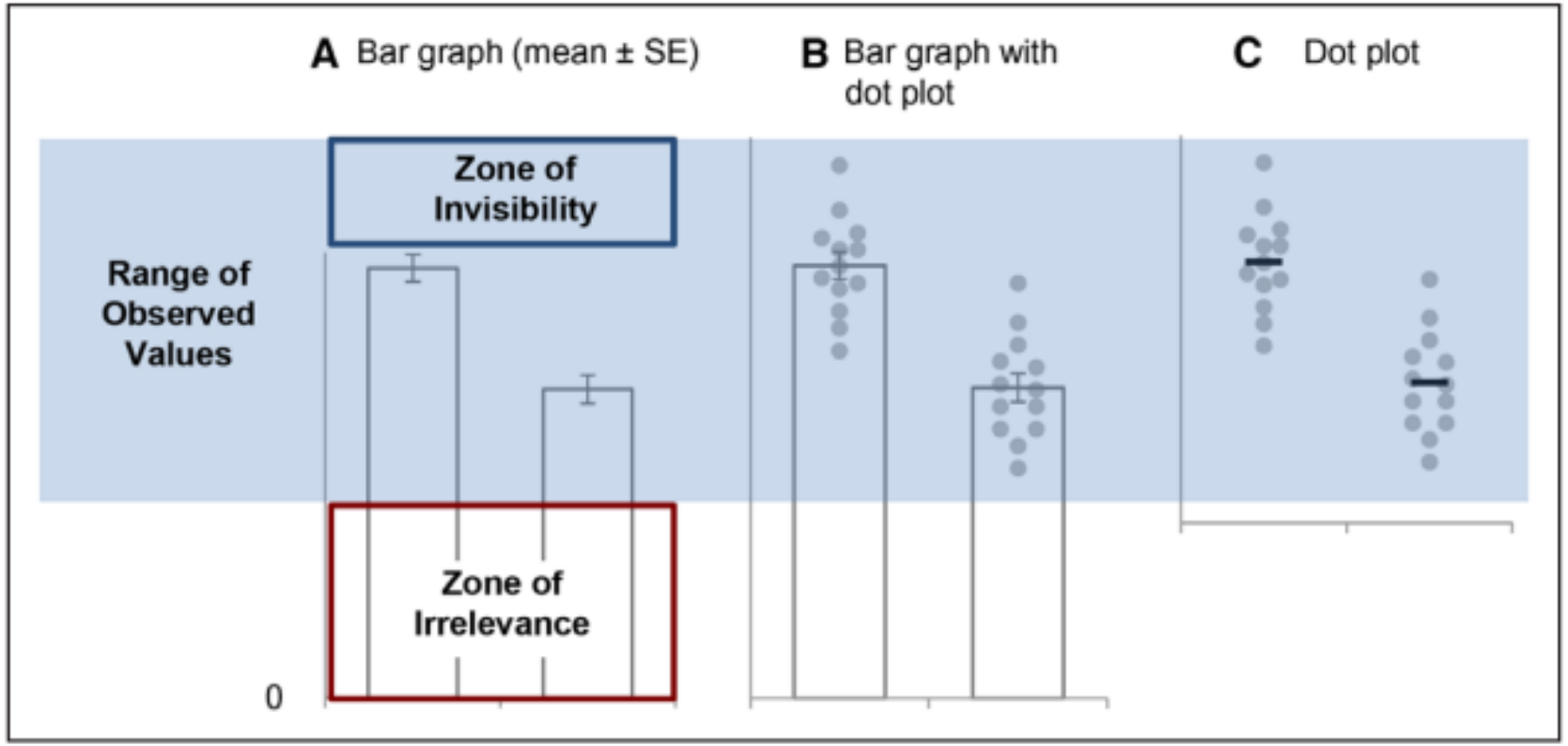


Figure 1. Why one shouldn't use a bar graph, even if the data are normally distributed.
Bar graphs arbitrarily assign importance to the height of the bar, rather than focusing attention on how the difference between means compares to the range of observed values. **A**, The bar height represents the mean. Error bars represent 1 standard error. The y-axis starts at zero and ends just above the highest error bar. **B**, Adding data points reveals that the bar graph in **A** includes low values that never occurred in the sample ("Zone of Irrelevance") and excludes observed values above the highest error bar ("Zone of Invisibility"). **C**, The dot plot emphasizes how the difference between means compares to the range of observed values. The y-axis includes all observed values. Reprinted from Weissgerber et al[22] under a CC-BY license. SE indicates standard error.

Weissberger et al. (2019)

We will discuss lots more later when we talk about data sheet and figure design.

Switch to ze code!